



Operating System

Lab 10

Fasiha Tariq – 53289

Submitted to:

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TASKS

1. Write a script that checks if a file named **data.txt** exists or not.

```
student@student-virtual-machine:~$ cat data.txt
Hello! Fasiha here
student@student-virtual-machine:~$ pico File
student@student-virtual-machine:~$ cat File
echo "Enter filename to check:"
read filename

    if [ -r "$filename" ]; then
        echo "File exist."
    else
        echo "File not exist."
    fi

student@student-virtual-machine:~$ chmod u+x File
student@student-virtual-machine:~$ ./File
Enter filename to check:
data.txt
File exist.
student@student-virtual-machine:~$ ./File
Enter filename to check:
Lab10.txt
File not exist.
```

2. Ask the user to enter a number and determine if it's even or odd.

```
student@student-virtual-machine:~$ pico even
student@student-virtual-machine:~$ cat even
#!/bin/bash

read -p "Enter a number: " num
result=$(expr $num % 2)
    if [ $result -eq 0 ]; then
        echo "$num is even "
    else
        echo "$num is odd"
    fi

student@student-virtual-machine:~$ chmod u+x even
student@student-virtual-machine:~$ ./even
Enter a number: 5
5 is odd
student@student-virtual-machine:~$ ./even
Enter a number: 8
8 is even
```

3. Ask the user for a score (0-100) and print the corresponding grade:

- 90+: A
- 80-89: B
- 70-79: C
- 60-69: D
- Below 60: F

```
student@student-virtual-machine:~$ pico score
student@student-virtual-machine:~$ cat score
#!/bin/bash
read -p "Enter a score (0-100): " score

if [ "$score" -ge 90 ]; then
    echo "Grade: A"
elif [ "$score" -ge 80 ]; then # 80-89
    echo "Grade: B"
elif [ "$score" -ge 70 ]; then # 70-79
    echo "Grade: C"
elif [ "$score" -ge 60 ]; then # 60-69
    echo "Grade: D"
else # Below 60
    echo "Grade: F"
fi
student@student-virtual-machine:~$ chmod u+x score
student@student-virtual-machine:~$ ./score
Enter a score (0-100): 70
Grade: C
student@student-virtual-machine:~$ ./score
Enter a score (0-100): 95
Grade: A
```

4. Take two numbers and an operator (+, -, *, /) from the user and perform the operation.

```
student@student-virtual-machine:~$ pico arithmeticop
student@student-virtual-machine:~$ cat arithmeticop
#!/bin/bash

read -p "Enter first number: " num1
read -p "Enter second number: " num2
read -p "Enter operator (+, -, *, /): " op

if [ "$op" = "+" ]; then
    result=$((num1 + num2))
elif [ "$op" = "-" ]; then
    result=$((num1 - num2))
elif [ "$op" = "*" ]; then
    result=$((num1 * num2))
elif [ "$op" = "/" ]; then
    result=$((num1 / num2))
else
    echo "Invalid operator."
    exit 1
fi

echo "Result: $result"
student@student-virtual-machine:~$ chmod u+x arithmeticop
student@student-virtual-machine:~$ ./arithmeticop
Enter first number: 5
Enter second number: 6
Enter operator (+, -, *, /): +
Result: 11
student@student-virtual-machine:~$ ./arithmeticop
Enter first number: 4
Enter second number: 5
Enter operator (+, -, *, /): -
Result: -1
student@student-virtual-machine:~$ ./arithmeticop
Enter first number: 7
Enter second number: 3
Enter operator (+, -, *, /): *
Result: 21
student@student-virtual-machine:~$ ./arithmeticop
Enter first number: 9
Enter second number: 4
Enter operator (+, -, *, /): /
Result: 2
```

